Attention alone does direct deduction: the rule and the iteration

Two registered counterfactual audits on the same three seeds of no_mlp L=16 d192, 350 cases each.

Q 51817042 | R 51820549 | commit d15ae24 | 3 seeds + 1 control | N=48 d5 q4 --shuffle-adj --fall | 350 cases/arm

The audit behind this programme had never touched this cell

- Findings 27/28 ran on N=24 full-architecture checkpoints only
- Stage Q edits a given hint; stage R edits a derived fact
- Same seeds, same instance stream, same 0.250 chance rate

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-57	`N0 N1 ... N47`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 393 TOKENS

HINTS	`BEGIN_HINTS N1 COLOR_2 N3 COLOR_1 N4 COLOR_3 N6 COLOR_3 N7 COLOR_3 N8 COLOR_0 ... (59 tokens)`
GRAPH	`BEGIN_GRAPH N0 N47 N3 N26 N33 N43 N1 N23 N32 N11 N12 N38 ... (289 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N5 COLOR_3 N11 COLOR_3 N14 COLOR_0 N16 COLOR_2 N17 COLOR_1 N23 ... (45 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 5 rounds, 19 deductions,
38 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000011, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

29 hints of 48 nodes, then 5 rounds of simultaneous deduction. Hints: N1=COLOR_2 N3=COLOR_1 N4=COLOR_3 N6=COLOR_3 N7=COLOR_3 N8=COLOR_0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N5 COLOR_3 N11 COLOR_3 N14 COLOR_0 N16 COLOR_2 ...
  11 deduced
BEGIN_ROUND          N18 COLOR_0 N25 COLOR_2 N35 COLOR_1 N47 COLOR_3
  4 deduced
BEGIN_ROUND          N0 COLOR_2 N19 COLOR_0
  2 deduced
BEGIN_ROUND          N46 COLOR_2
  1 deduced
... 1 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N5, neighbours N2 N15 N9 N13 N36

their colours known before this round:

```
   [0, 1, 2]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N2
```

only colour left: 3

```
so the trace emits  N5 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the distribution both audits drew their cases from.

T=5 rounds, the median of 40 draws from this config (range 1-19) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

One hint changes, and the model's answer changes with it

Stage Q's case #1, regenerated from the arm's own --seed-start and matched against row 1 of its behaviour CSV before drawing.

THE EDIT: ONE TOKEN OF 391

before	N0	COLOR_3	N1	COLOR_3	N2	**COLOR_2**	N3	COLOR_1	N4	COLOR_1	N5
after	N0	COLOR_3	N1	COLOR_3	N2	**COLOR_0**	N3	COLOR_1	N4	COLOR_1	N5

position 6, in the HINTS section

N2 is a HINT: a fact the model was GIVEN.

WHY DD'S ANSWER CHANGES

node N9, deduced in round 0

neighbours known before: N2=2 N4=1 N28=3 N36=1 N43=3

covered colours [1, 2, 3] = q-1 of 4

so N9 must be COLOR_0

recolour N2 to COLOR_0 and
COLOR_2 is the only one left:

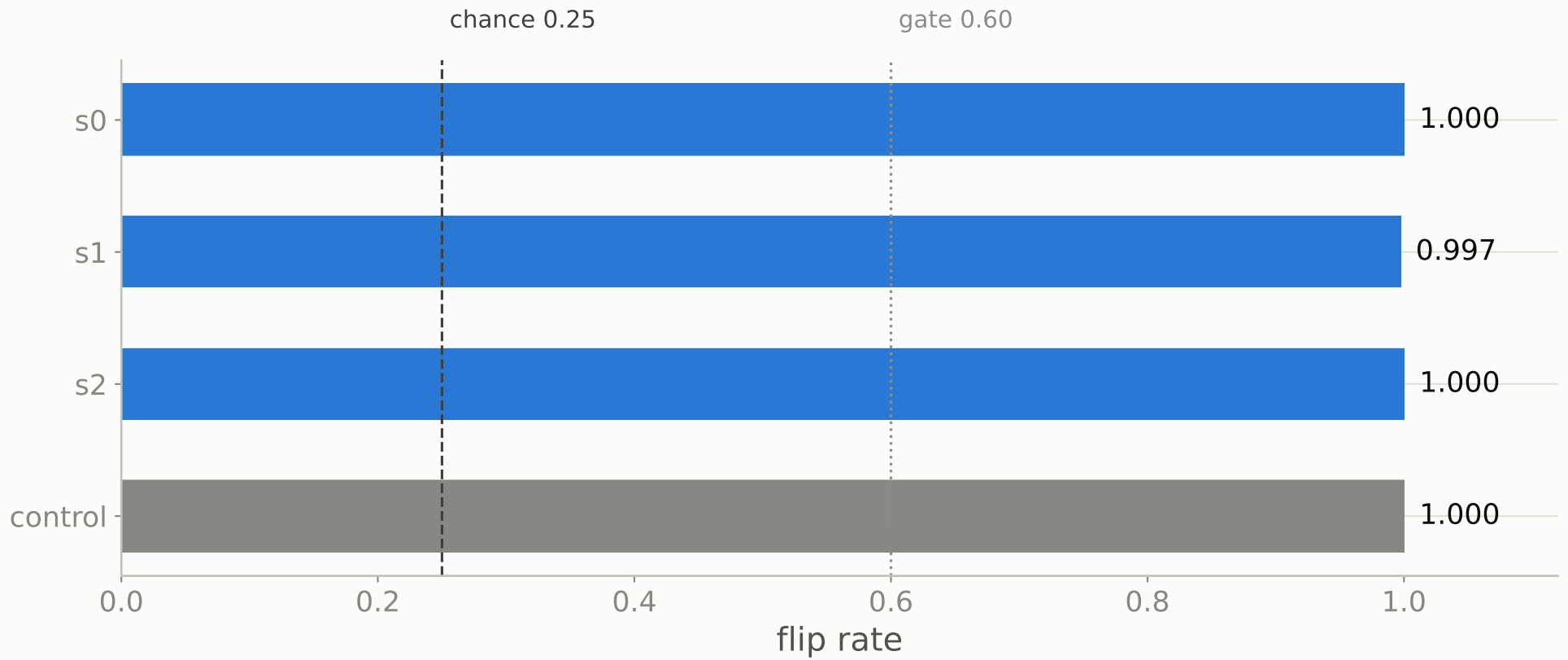
DD's answer becomes COLOR_2

WHAT THE MODEL DID ON THIS CASE (row 1 of Q_noff_s0's behaviour CSV, not recomputed here)

before the edit	**predicted COLOR_0**	**CORRECT**	logit(freed) - logit(true) -14.69 -> +16.12
after the edit	**predicted COLOR_2**	**FOLLOWED THE EDIT**	*the sign reverses: the edit moves it from confidently the true colour to confidently the freed one*

Round-0 forcing: every known neighbour of N9 is a HINT, so this case tests the elimination RULE over GIVEN facts. The model is never shown the reasoning on the right -- only the node/colour pairs.

Editing one hint flips the answer on essentially every case



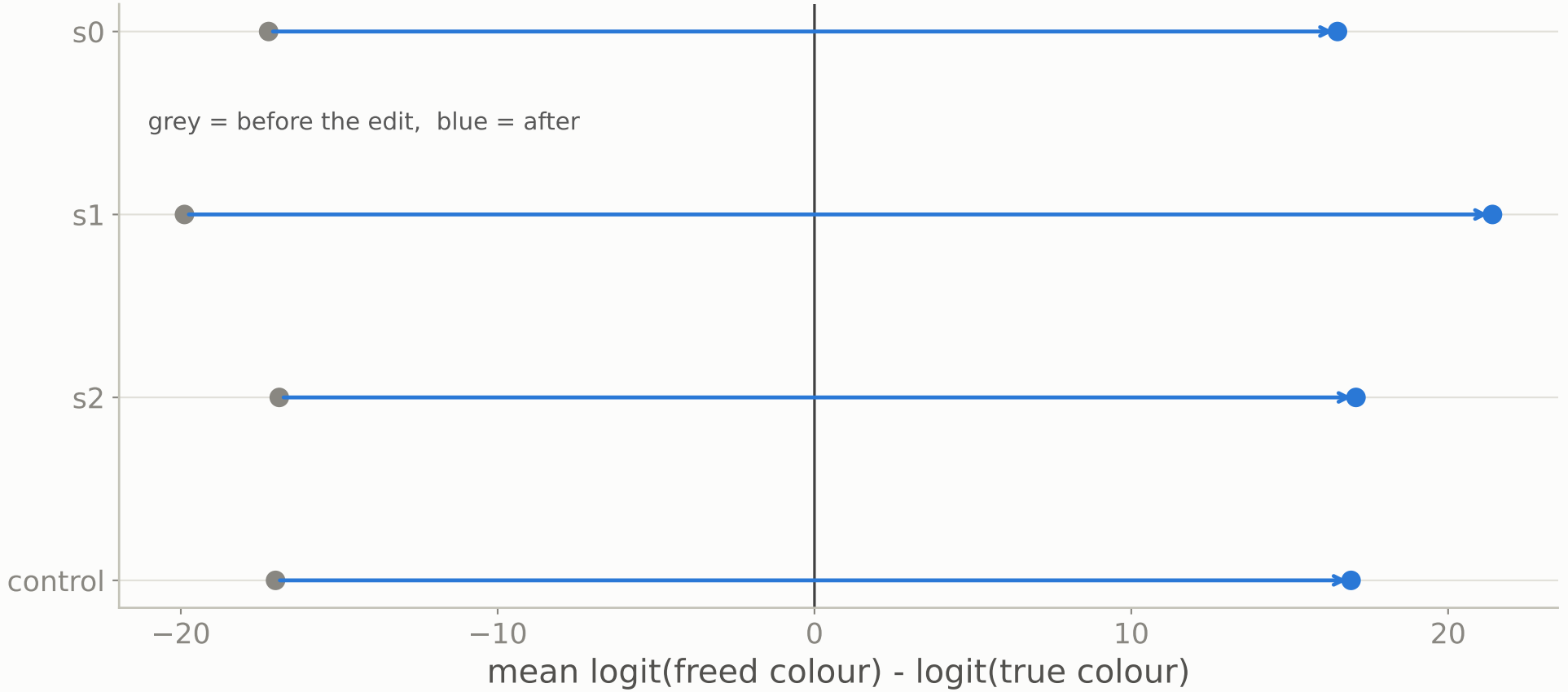
350 counterfactuals per arm, scored against absolute chance. The control is a different cell in two respects.

Before the edit, every arm is correct on all 350 cases

arm	clean	flipped	x chance	logit gap
s0	1.000	1.000	4.00	-17.2 to +16.5
s1	1.000	0.997	3.99	-19.9 to +21.4
s2	1.000	1.000	4.00	-16.9 to +17.1
control	1.000	1.000	4.00	-17.0 to +16.9

Q3's precondition is clean >= 0.50. Every arm sits at the ceiling, which is what Q2 and R2 are about.

The logit gap reverses sign, from -17.2 to +16.5



Not a marginal preference flip. Magnitudes are not comparable across arms of different width and are not compared.

Stage Q tested round zero only, where every neighbour is a hint

- find_flippable_deductions picks nodes forced in round 0
- clean_correct is 1.000 because these cases are easy by construction
- A one-step classifier would score exactly the same here

Q4's registered seed rule fired on noise and means nothing

- A 0.003 seed spread against a 0.001 arm-control gap
- Every arm saturated, so the comparison has no resolution
- Stage R registers a 0.95 ceiling rule so this cannot recur

Stage R edits a colour the model itself deduced earlier

Stage R's case #1, same check against row 1 of its CSV. The edited token is a previously EMITTED DEDUCTION, not a hint.

THE EDIT: ONE TOKEN OF 391

before	COLOR_1	N22	COLOR_2	N23	**COLOR_2**	N24	COLOR_3	N39	COLOR_0
after	COLOR_1	N22	COLOR_2	N23	**COLOR_3**	N24	COLOR_3	N39	COLOR_0

position 377, in the REASONING section

N23 was DEDUCED by the model in round 0: a fact it DERIVED.

WHY DD'S ANSWER CHANGES

node N16, deduced in round 1

neighbours known before: N11=1 N23=2 N33=0 N34=1

covered colours [0, 1, 2] = q-1 of 4

so N16 must be COLOR_3

recolour N23 to COLOR_3 and

COLOR_2 is the only one left:

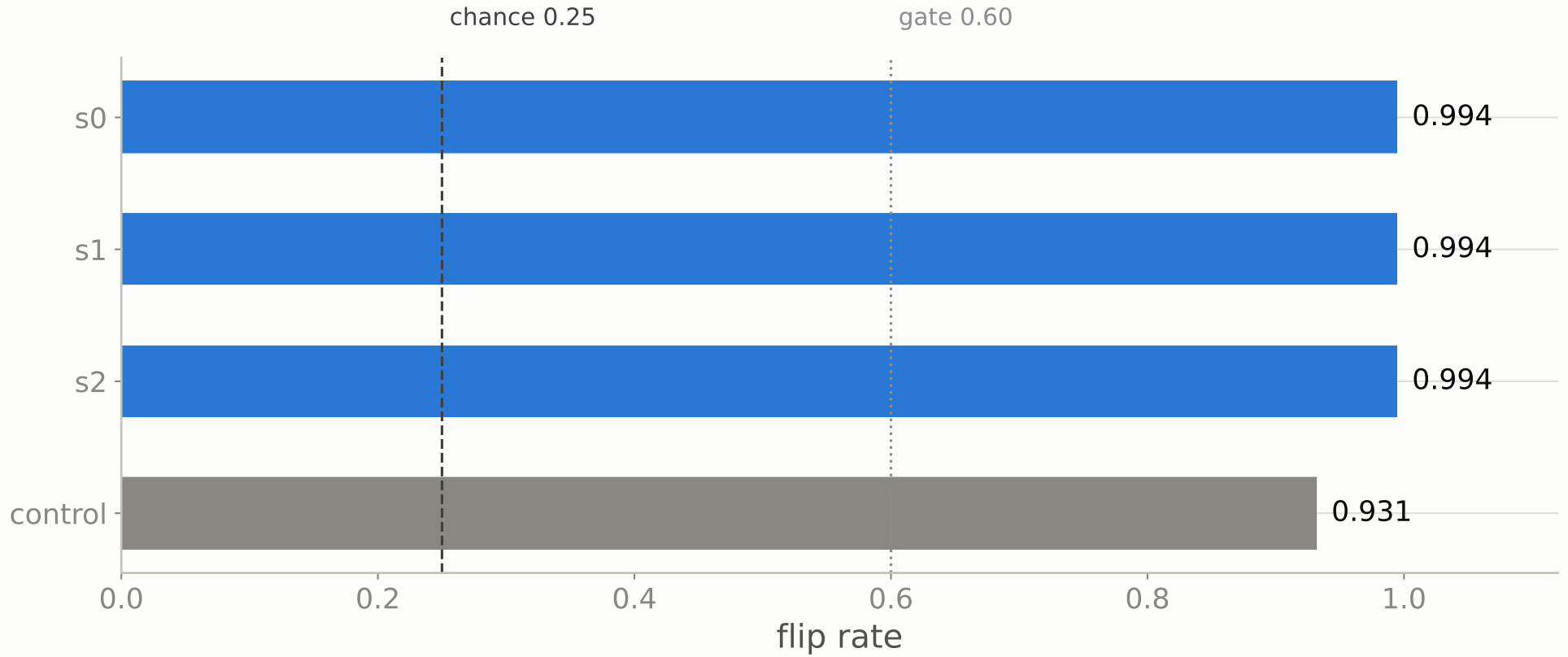
DD's answer becomes COLOR_2

WHAT THE MODEL DID ON THIS CASE (row 1 of R_noff_s0's behaviour CSV, not recomputed here)

before the edit	**predicted COLOR_3 CORRECT**	logit(freed) - logit(true) -14.32 -> +11.64
after the edit	**predicted COLOR_2 FOLLOWED THE EDIT**	*the sign reverses: the edit moves it from confidently the true colour to confidently the freed one*

This is the difference between applying an inference rule once and running the algorithm: N23's colour was derived in round 0, and N16 is forced in round 1 against it.

Editing a derived fact flips the answer 3.98 times above chance



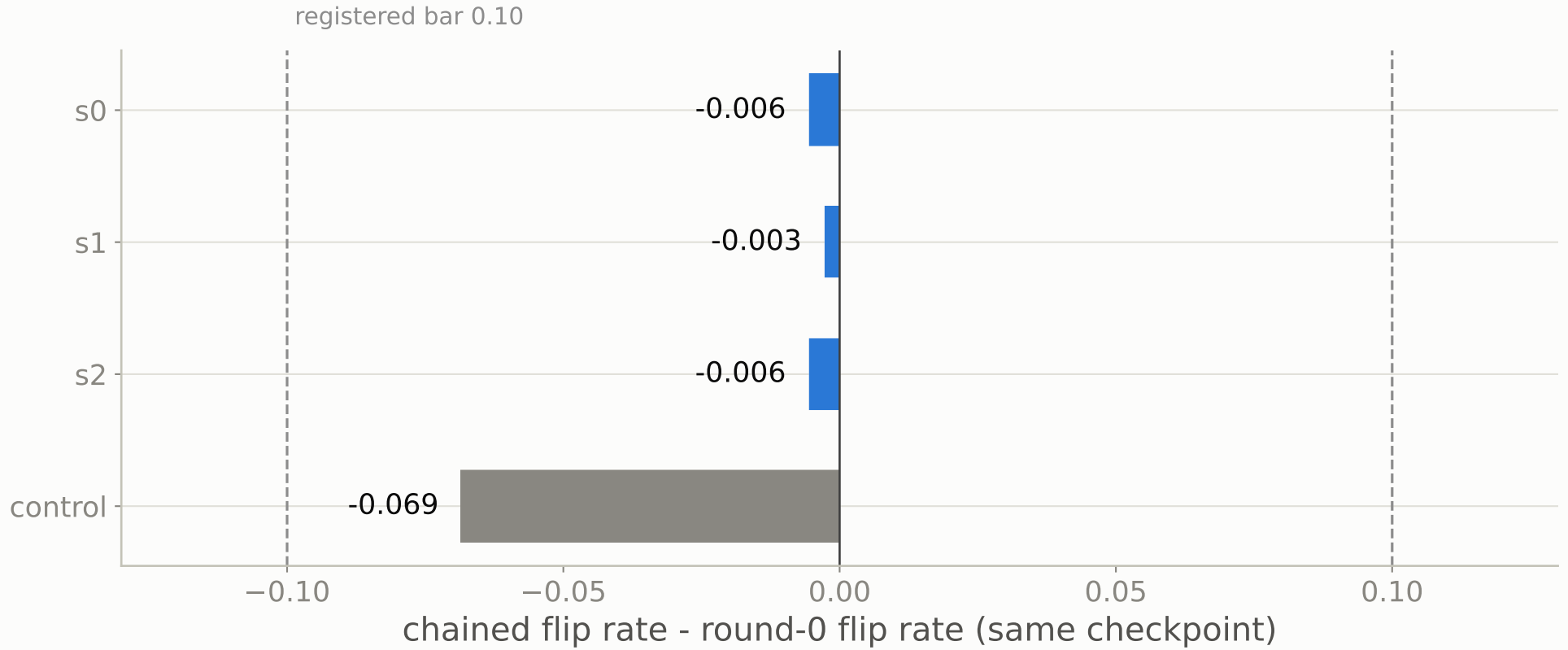
R0's gate was 0.60 on two of three seeds, registered before the run. It held on all three.

Chained cases are harder, yet clean accuracy is still 1.000

arm	clean	flipped	reverts	logit gap
s0	1.000	0.994	0.006	-16.2 to +15.6
s1	1.000	0.994	0.000	-18.4 to +16.8
s2	1.000	0.994	0.003	-18.0 to +16.8
control	1.000	0.931	0.043	-19.0 to +7.8

reverts_to_original is R1's detection signature: the model keeping its own answer rather than failing to chain.

Using a derived fact instead of a given one costs nothing



R4, the one licensed cross-stage subtraction: same checkpoint, same instance stream, disjoint case sets by construction.

The three seeds' identical 0.994 is a coincidence of counts

seed	flipped	misses	which cases
s0	0.994	2 of 350	N42@r1 N42@r1
s1	0.994	2 of 350	N29@r1 N10@r1
s2	0.994	2 of 350	N41@r1 N42@r1

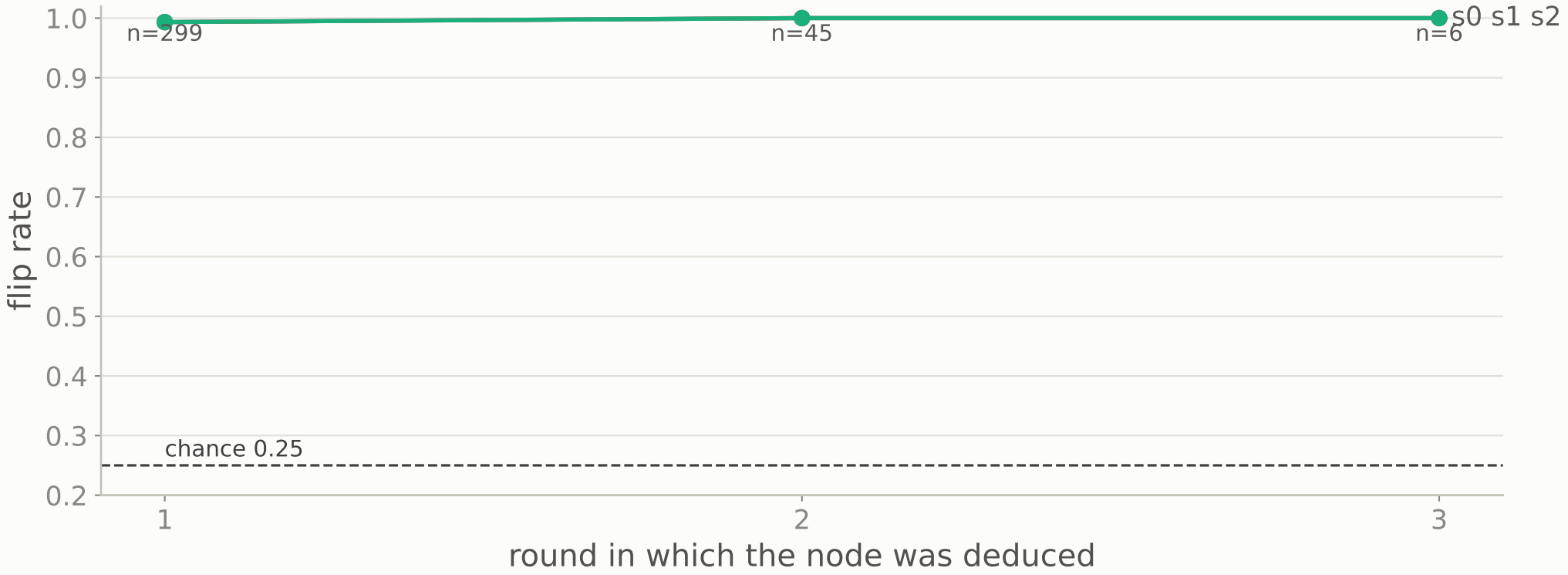
Node and round are not an instance id: s0 missed node 42 twice, in different instances. No two seeds miss the same pair.

The control's lower flip rate is detection, not failure to chain

arm	flipped	misses	of which revert
s0	0.994	2	2
s1	0.994	2	0
s2	0.994	2	1
control	**0.931**	**24**	**15**

The full solver is the one more likely to notice the prefix was edited and keep its own value. Shape-confounded either way.

The flip rate does not decay with the round being deduced



Composing layers predicts no decay, and none appears. Monotone decay would have tied rounds to layers quantitatively.

The chained cases reach round three, so depth is barely probed

- 299 of 350 cases are round 1, 6 are round 3
- No decay is therefore tested to 3 of 16 layers
- Rounds beyond the third are untested on this cell

The guard third of the original audit is still untested here

- 27/28 also found 0 of 350 emissions when forcing broke
- No module in the current tree tests that on no_mlp
- So declining to deduce what does not follow stays unmeasured

No previous audit of this cell exists on this eval

quantity	findings 27/28	stages Q+R
graph size	N=24	N=48
architecture	full	no_mlp L=16 d192
round-0 flip	350 of 350	1.000, 0.997, 1.000
chained flip	0.980	0.994, 0.994, 0.994
seeds	1	3

Different N and different architecture, so these are not comparable numbers -- they are the reason the audit was rerun.